

FORT SHEVCHENKO, Kazakhstan

WMO: 380010

Lat: 44.5476N

Lon: 50.2425E

Elev: -25

StdP: 101.63

Time Zone: 5.00 (E05)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.0	-11.2	-19.8	0.6	-12.8	-16.8	0.9	-8.9	15.5	-0.4	14.1	-0.6	5.1	50	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	37.2	18.9	35.8	18.7	33.9	18.7	25.7	29.3	24.4	28.2	23.3	27.5	5.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	19.6	28.2	23.2	17.9	27.2	21.9	16.5	26.3	79.1	29.1	73.8	28.0	69.3	27.4	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	12.1	10.3	DB	-17.4	40.1	5.0	1.4	-21.0	41.1	-23.9	41.9	-26.7	42.7	-30.3	43.7
			WB	-17.0	26.9	5.3	1.7	-20.9	28.1	-24.0	29.0	-27.0	30.0	-30.9	31.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	-1.2	0.2	5.4	11.8	18.7	24.2	27.1	26.3	20.3	13.0	5.9	0.9	(6)
(7)		DBStd	10.83	4.89	4.89	3.72	3.99	4.10	3.66	2.98	3.40	3.79	4.23	4.47	4.28	(7)
(8)		HDD10.0	1230	347	276	149	24	0	0	0	0	20	133	281		(8)
(9)		HDD18.3	2867	605	509	402	198	46	2	0	1	22	171	373	539	(9)
(10)		CDD10.0	2250	0	0	6	78	271	427	531	507	308	112	11	0	(10)
(11)		CDD18.3	847	0	0	0	3	58	179	273	249	80	5	0	0	(11)
(12)		CDH23.3	9197	0	0	1	27	508	1896	3209	2867	652	38	0	0	(12)
(13)		CDH26.7	4254	0	0	0	4	176	887	1561	1398	225	4	0	0	(13)
(14)	Wind	WSAvg	5.1	5.3	5.4	5.6	5.5	4.8	5.1	4.7	4.6	4.9	5.0	5.1	5.5	(14)
(15)	Precipitation	PrecAvg	143	6	7	13	16	16	12	10	7	12	10	14	12	(15)
(16)		PrecMax	337	23	28	43	67	113	87	51	56	76	31	52	36	(16)
(17)		PrecMin	70	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	63	6	7	12	17	23	18	12	12	15	8	12	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	12.5	18.9	26.2	34.1	38.0	39.1	39.2	34.7	26.8	18.2	11.9	(19)
(20)		0.4%	MCWB	7.1	8.5	10.6	14.5	16.7	18.6	19.8	19.3	18.5	15.8	11.5	8.4	(20)
(21)		2%	DB	7.2	9.3	15.2	23.0	31.0	35.9	37.2	37.1	31.7	24.1	15.4	9.3	(21)
(22)		2%	MCWB	5.8	6.2	8.7	13.5	15.9	17.9	19.1	19.1	16.5	14.2	11.5	6.8	(22)
(23)		5%	DB	6.0	7.7	12.9	20.7	28.2	33.9	35.8	35.2	29.1	21.9	14.0	7.9	(23)
(24)		5%	MCWB	4.6	5.3	7.9	12.9	15.8	17.8	19.0	18.8	16.2	14.1	10.8	6.0	(24)
(25)		10%	DB	4.8	6.1	10.8	18.2	26.0	31.9	33.8	33.1	27.0	19.8	12.2	6.4	(25)
(26)		10%	MCWB	3.5	4.2	7.0	11.9	15.8	17.8	19.4	19.0	16.2	13.8	9.6	4.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	8.9	11.6	17.1	20.9	24.9	27.5	27.9	23.2	19.7	14.1	9.2	(27)
(28)		0.4%	MCDWB	8.9	11.9	16.5	21.5	25.7	28.0	30.9	31.5	26.8	22.6	16.1	10.8	(28)
(29)		2%	WB	6.0	6.8	9.9	15.1	19.3	23.3	25.8	25.9	21.4	17.5	12.6	7.4	(29)
(30)		2%	MCDWB	7.1	8.6	13.9	20.1	24.1	26.8	29.4	29.6	25.5	20.5	14.4	8.9	(30)
(31)		5%	WB	4.6	5.6	8.6	13.9	18.3	22.2	24.7	24.5	20.2	16.1	11.3	6.1	(31)
(32)		5%	MCDWB	5.7	7.1	11.8	18.4	23.2	26.2	28.3	28.4	24.3	19.3	13.3	7.4	(32)
(33)		10%	WB	3.4	4.4	7.4	12.8	17.4	21.1	23.6	23.1	19.1	14.8	9.9	4.8	(33)
(34)		10%	MCDWB	4.4	5.7	10.0	17.1	22.6	25.7	27.7	27.6	23.5	17.9	12.2	6.1	(34)
(35)	Mean Daily Temperature Range	MDBR	7.0	7.9	8.4	9.6	10.4	11.2	11.0	11.6	11.0	10.0	8.1	6.8	(35)	
(36)		5% DB	MCDBR	7.2	9.6	12.2	13.5	14.4	14.7	14.4	15.2	14.3	13.4	10.5	8.4	(36)
(37)		5% DB	MCWBR	6.2	7.3	7.3	7.1	7.0	7.2	7.1	7.7	8.1	8.3	7.7	6.7	(37)
(38)		5% WB	MCDBR	7.0	8.5	10.8	11.2	11.3	10.7	9.9	10.7	10.9	9.9	9.3	7.3	(38)
(39)		5% WB	MCWBR	6.2	6.9	7.2	6.6	6.7	7.1	7.2	7.6	8.3	8.0	7.5	6.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.393	0.413	0.448	0.517	0.497	0.483	0.492	0.491	0.458	0.422	0.399	0.405	(40)	
(41)		taud	2.107	2.072	2.037	1.900	2.020	2.109	2.105	2.082	2.112	2.167	2.142	2.069	(41)	
(42)		Ebn at Noon	674	741	772	751	784	796	783	766	755	725	662	607	(42)	
(43)		Edh at Noon	102	125	148	184	169	155	154	152	136	113	97	97	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.25	2.21	3.51	5.00	6.31	7.06	6.73	6.06	4.67	2.81	1.56	0.92	(44)	
(45)		RadStd	0.13	0.28	0.34	0.46	0.44	0.34	0.34	0.25	0.31	0.30	0.17	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-1.67	-2.15	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page